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Makeshift USB - SATA Adapter

Putting old stuff to use, the DIY way

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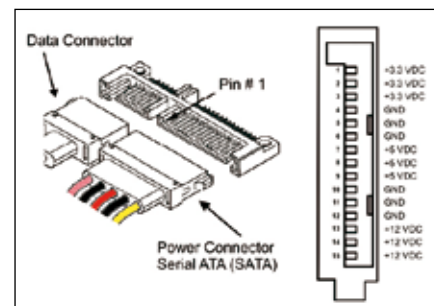
Don't throw away your spoilt external drive yet, for it could be of some use to you. These external drives contain a simple USB-SATA bridge IC which can be adopted to handle all kinds of drives if you have the right power source.

To see if your old drive is worth anything, first split open your external storage drive enclosure. You'll come across a hard drive encased in a rubberised protective sheath. Check the SATA interface of the

We are interested in the power connector (shown on the right). We'll have to make a power source so that all sorts of hard drives can be powered while we hook up a normal USB cable to the adapter to run it. 3.5-inch hard drives draw power primarily from the 12 volt and 5 volts lines while the 2.5-inch hard drives draw power from the 5 volt line with the 12 volt line used in rare cases for the initial drive spin-up. The power consumption chart tells us that we'll need a maximum of 1 ampere on the 12 volt line and about 1.2 amperes on the 5 volt line.

So basically, we'll be having our power source (with a current limiter) which is connected to the hard drive, the hard drive in turn is hooked onto the adapter using a MALE-FEMALE SATA data cable. A normal laptop charger can be used as a DC power source for obtaining the two power lines which we'll need. We can use two linear power regulators to get the job done, however, it would be prudent to limit the input current for both the regulators so that accidental burnouts can be avoided. For this we'll be using a simple current limiter circuit. The reason being that in case there is an electrical short in your circuit then you'll end up blowing up your power source by either melting your fuse or transformer winding. Since we don't want that happening to your laptop charger, the current limiter was added to the circuit.

The 5 volt line can be sourced from a normal USB Y cable that has two USB inputs to get ample current from the computer. Warning! Do not source without a current limiter circuit or you



SATA power connector pinouts

may cause irreparable damage to the computer's motherboard. While for the 12 volt line we'll be using the 7812 linear voltage regulator.

Most laptop chargers are rated at 65 watts and output 19 volts @ 3-4 amperes. We'll be drawing out 12 volts from this. This is a simple current limiter circuit that uses two NPN transistors to limit the current on the load line to 1 Ampere. R2 is a current sense resistor which monitors the current going through T1. The voltage across R2 increases as current through T1 rises. If the voltage on the top of R2 reaches the cutoff limit then T2 is turned on which in turn channels some of the current into the ground. Here is a circuit simulator java applet [<http://dgit.in/RJc6fJ>]. You can vary the hfe values of the NPN transistors to get the proper output as you desire. In our case it is 1 Ampere. Ensure the resistors (R1 and R2) are of proper wattage on all lines or you'll risk burning them out.

The linear voltage regulator 7812 is then placed across the 19V (laptop input) and collector of the transistor T1. Remember to select a 7812 that can handle



Left - Separate PCBs, Right - Integrated PCB

drive, you'll either have a hard drive that has a USB bridge IC (JM20329 in this case) on the HDD PCB or it'll be separate.

What we want is for the hard drive to be separated altogether from the hard drive PCB like the one in the left-hand portion of the image. At this moment you can put in a new hard drive and close the enclosure to use it as a normal external like you have been doing for so long, or you could go through the following steps to adapt it for multiple hard drive types. However, if your PCB itself is messed up then there is nothing you can do than throw it away. You could hook up your drive to see if it is still working.

The SATA interface consists of 2 ports, the 7-pin data port and the 15-pin power port.

Power consumption of hard drives (Approximate values)

Average Power Consumption (Watts)	12v line	5v line
2.5" SSD	-	3 W
5400 RPM 2.5" HDD	-	6 W
7200 RPM 2.5" HDD	-	6.8 W
Green 3.5" HDD	3.74 W	1.36 W
7200 RPM 3.5" HDD	10.8 W	2.5 W
10000 RPM 3.5" HDD	10 W	3.2 W
15000 RPM 3.5" HDD	12.7 W	5.4 W